`dot(dot(X,Y),Z) → dot(X,dot(Y,Z))`

A short equation, but replacement dots need new attributes.

Schematic code
g: graph; m: metadata

Traditional approaches

(a) Algebraic term

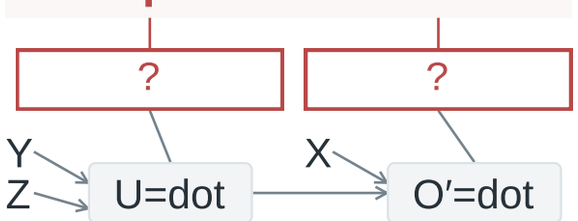
Pattern

```
(dot (dot X Y) Z)
=> (dot X
     (dot Y Z))
```

Supplies the RHS graph, but leaves both dot attributes unspecified.

Attribute handling missing

Incomplete tensor rewrite



Concise rule: ✓

Attributes: ✗

(b) Custom logic

Matching / conditions

```
match_nested_dots(g);
check_conditions(m);
```

Dimension mapping

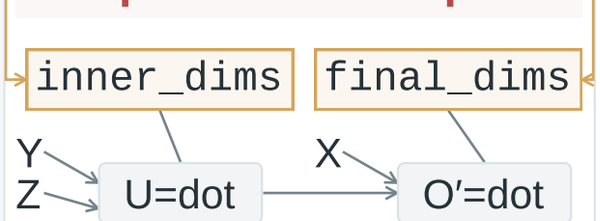
```
inner_dims =
compute_inner_dims(m);
final_dims =
compute_final_dims(m);
```

RHS construction

```
U=build_dot(Y,Z,
  inner_dims);
return build_dot(X,U,
  final_dims);
```

Manual code implementation

Complex and error-prone



Concise rule: ✗

Attributes: ✓ Manual

TEPL

(c) Explicit attributes

Rule

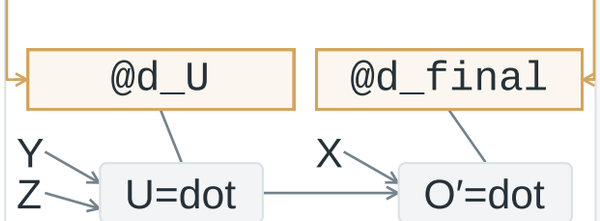
```
(dot[@outer]
 (dot[@inner] X Y) Z)
=> (dot[@d_final] X
     (dot[@d_U] Y Z))
```

User-written attribute expressions

```
derive {
 @d_U = f_U(m);
 @d_final=f_final(m);
}
```

Schematic dimension logic

User-written expressions
m: source metadata



Concise rule: ✓

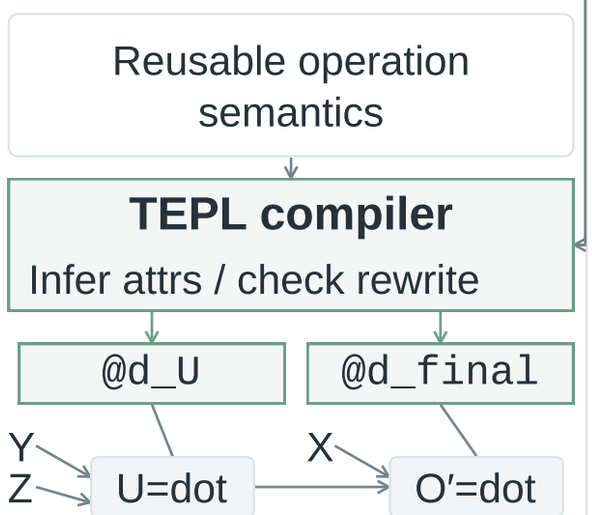
Attributes: ✓ Manual

(d) Inferred attributes

Rule

```
(dot (dot X Y) Z)
=> (dot[auto] X
     (dot[auto]
       Y Z))
```

Each **auto** denotes an independent function of source shapes and attributes.



Concise rule: ✓

Attributes: ✓ Automatic